Week 4 Day 4: Tuning Report

Model Tuning, Regularization & Reproducible Pipelines

Hyperparameter Search

The two best models from the previous day, **Logistic Regression** and **LightGBM**, were tuned using RandomizedSearchCV with 5-fold Stratified Cross-Validation. Since my primary evaluation metric is **Precision**, the search was optimized using Precision instead of Accuracy or F1.

For Logistic Regression, the best parameters were L1 regularization with C = 0.001. This showed that a simpler model with stronger regularization performed better than a more complex one.

For LightGBM, the best parameters were learning_rate = 0.01, max_depth = 3, n_estimators = 100, and l2_regularization = 1.0. These settings produced the highest cross-validated Precision while keeping the model well controlled and reducing overfitting.

	Model	Best Precision
0	Logistic Regression	0.769856
1	LightGBM	0.981490

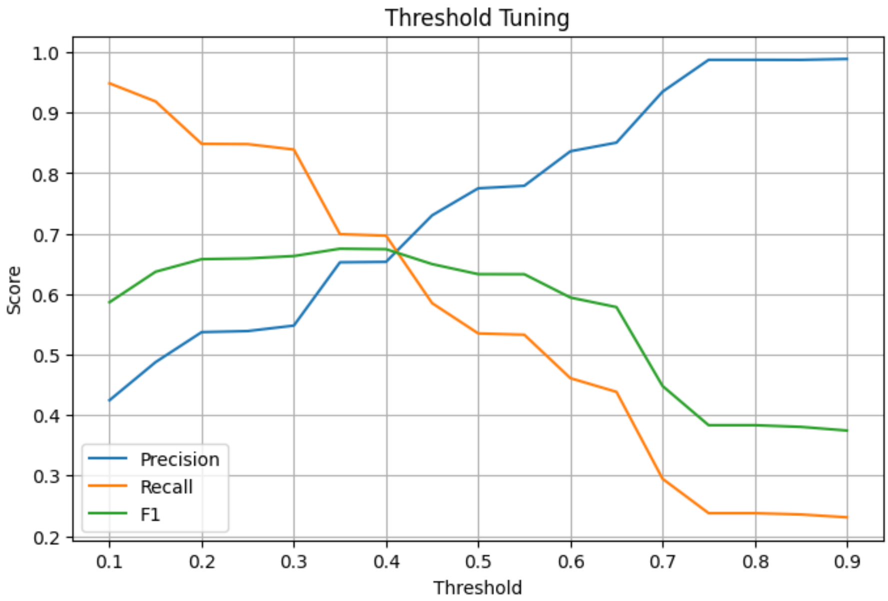
Model Diagnostics

Learning curves were used to check whether the models were overfitting or underfitting. Both Logistic Regression and LightGBM showed training and validation curves that stayed close together and leveled off as more training data was added. This indicates **low variance**, meaning the models generalize well without overfitting. Since both models also reached consistently high Precision, they show **low bias** and are able to learn the important patterns in the data.

The effect of **C** on Logistic Regression showed that stronger regularization gave the best validation performance, while larger values of C slightly reduced generalization. For LightGBM, increasing **max_depth** beyond **5** improved the training score but reduced validation Precision, showing the start of overfitting. Keeping the trees shallow was therefore the better choice.

Probability calibration was then evaluated using calibration plots and the **Brier Score**. The original model achieved a **Brier Score of 0.1191**, indicating that its predicted probabilities were already reasonably reliable. After comparing **Sigmoid** and **Isotonic** calibration, Isotonic produced a calibration curve that stayed closer to the ideal diagonal, so it was selected as the final calibration method.

Different classification thresholds were also tested. Lower thresholds increased Recall but also produced more false positives, while higher thresholds greatly improved Precision at the cost of Recall. Since the goal of this project is to maximize Precision, a threshold of **0.75** was selected. At this threshold, the model achieved **98.7% Precision**, meaning almost every predicted high-income individual was actually correct.



Final Model Performance

The final model is a **tuned and isotonic-calibrated LightGBM pipeline** using a classification threshold of **0.75**. It achieved the following results on the untouched hold-out test set:

	Metric	Score
0	Accuracy	0.816870
1	Precision	0.987567
2	Recall	0.237810
3	F1	0.383316
4	ROC AUC	0.895478

The confusion matrix showed very few false positives (7), which matches the project's main objective of maximizing Precision. Although the model misses a number of actual high-income individuals, this trade-off is acceptable because making incorrect positive predictions is considered more costly.

Conclusion

Overall, tuning helped improve the final model and made its predictions more reliable. **LightGBM** stayed the best model after tuning, and isotonic calibration made its probability estimates more accurate. Precision was maximized (**98.7%**) which was my main goal from the start. The final pipeline is saved, reproducible, and ready to use on new data! Thank you for reading